

Numerical Methods Question

- What is the mathematical assumption being made for finding a real root of a non-linear equation? Find a real root of the equation:
 $\cos(x) + e^x + x^2 = 3$ by using Bisection method up to 3 decimals.
- Use Newton-Raphson method to approximate the root of the function $f(x) = x^4 - 4x^3 + 4x^2 + x - 4$ with error less than 0.0001
- Write an algorithm and pseudocode of fixed-point iterative method.
- The following table gives the displacement, x(cms) of an object at various time, t(seconds). Find the displacement of this object at t=1.3 second and t = 1.5 second, using any suitable interpolation formula.

t	1.0	1.2	1.4	1.7
x	9.0	9.5	10.2	11.0

- Estimate the value of $\sin\theta$ at $\theta = 25$ using Newtons forward interpolation.

θ	10	20	30	40	50
$\sin\theta$	0.1736	0.3420	0.5	0.6428	0.7660

- Find the cubic spline interpolation formula for the following data:

x	1	2	3	4	5
f(x)	1	0	1	0	1

- Apply Lagrange's method to find the value of x when $f(x) = 15$ from the given data

x	5	6	9	11
f(x)	12	13	14	16

- The growth of bacteria (N) in a culture after t hours is given by the following table.

Time t(hr)	0	1	2	3	4
Bacteria (N)	32	47	65	92	132

If the relationship between bacteria N and time t is of the form $N = ab^t$. Using least square approximation estimate the N at t = 5hr.

- Integrate the following function by using Simpson's 3/8 rule.

$$\int_0^{\pi/2} \frac{\sin x}{x} dx$$

- Find the first and second derivatives of f(x) at x = 1.5 if

x	1.5	2.0	2.5	3.0	3.5	4.0
f(x)	3.375	7.000	13.625	24.000	38.875	59.000

- Use Romberg integration to evaluate $\int_0^{3\pi/2} e^x \sin x dx$.

12. Explain an application of Numerical Methods in Engineering and Error in Numerical Method.